

Analysis Details

Pod Information

Namespace	Pod Name	Last Recorded Phase
chaos-test	heap-oom-prom-75d49588db-9995t	Running
Container Status		
heap-oom-prom	NOT READY Restart Count: 6 Last Recorded State: Waiting(CrashLoopBackOff) Last Terminated: Error (exit 134)	

Root Cause Analysis

Root Cause Analysis of OutOfMemoryError in Heap-Oom-Prom Container

The container experienced an OutOfMemoryError due to excessive memory consumption, which was caused by a combination of factors including high MEM usage and insufficient memory allocation. The logs show multiple warnings and errors related to OutOfMemoryError, indicating that the issue is persistent and not temporary.

Evidence

Metrics: The pod's memory utilization is consistently above 90% (511MB/512MB), indicating that there is not enough memory available for the container to function properly. Logs: The logs show multiple warnings and errors related to OutOfMemoryError, including a fatal error message 'BackOff: Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-75d49588db-9995t_chaos-test(84bc20a4-15ab-416c-b290-328291109dbf),'. Events: The events log shows multiple entries related to scheduling and restarting the container, indicating that the container is repeatedly experiencing OutOfMemoryError.

Supported Logs

View Logs

- BackOff: Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-75d49588db-9995t_chaos-test(84bc20a4-15ab-416c-b290-328291109dbf),
- BackOff: Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-67d785cb84-f8zfd_chaos-test(bb58d754-4b03-4488-9d11-c106ef708edf),
- BackOff: Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-67d785cb84-d7vj_l_chaos-test(18f6bb00-524a-4f2d-88b8-0aa6d47782a9)

Validation Summary

OVERALL ASSESSMENT

The RCA evidence strongly supports the claim that the container experienced an OutOfMemoryError due to excessive memory consumption. The logs and events show multiple warnings and errors related to OutOfMemoryError, indicating a persistent issue.

VALIDATION TRACE

- The container experienced an OutOfMemoryError due to excessive memory consumption Supported 70%

Evidence:
BackOff: Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-75d49588db-9995t_chaos-test(84bc20a4-15ab-416c-b290-328291109dbf),
BackOff: Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-67d785cb84-f8zfd_chaos-test(bb58d754-4b03-4488-9d11-c106ef708edf),
BackOff: Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-67d785cb84-d7vj_l_chaos-test(18f6bb00-524a-4f2d-88b8-0aa6d47782a9)

Analysis Metadata

STATUS

Completed

STARTED

2026-03-03 09:49:03.497

DURATION

23m 13s

TOKEN COUNT (USED FOR CONTEXT)

1079

Stage Performance

DATA COLLECTION 0s	ANOMALY DETECTION 6m 35s	RCA ANALYSIS 7m 32s	VALIDATION 9m 4s
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Context Details

Metrics

MEMORY USAGE	MEMORY LIMIT	MEMORY %	CPU USAGE	CPU LIMIT	CPU %
535855104.00 bytes	536870912.00 bytes	99.81%	0.045 cores	0.500 cores	9.03%

LLM SUMMARY: mem=511MB/512MB (99.8%) cpu=0.045c/0.500c (9.0%)

Events

29 raw / 29 dedup

NORMAL Scheduled

x1

2026-03-03T08:16:43Z

Successfully assigned chaos-test/heap-oom-prom-67d785cb84-d7vj1 to ip-10-0-2-33.us-east-2.compute.internal

NORMAL AddedInterface

x1

2026-03-03T08:16:44Z

Add eth0 [10.129.2.15/23] from ovn-kubernetes

NORMAL Pulling

x1

2026-03-03T08:16:44Z

Pulling image "quay.io/causa-ai-hub/quarkus-heap-oom:heap-oom-prom"

NORMAL Pulled

x1

2026-03-03T08:16:55Z

Successfully pulled image "quay.io/causa-ai-hub/quarkus-heap-oom:heap-oom-prom" in 10.737s (10.737s including waiting). Image size: 336772927 bytes.

NORMAL Created

x9

2026-03-03T08:43:03Z

Created container: heap-oom-prom

NORMAL Started

x9

2026-03-03T08:43:03Z

Started container heap-oom-prom

NORMAL Pulled

x8

2026-03-03T08:43:03Z

Container image "quay.io/causa-ai-hub/quarkus-heap-oom:heap-oom-prom" already present on machine

WARNING BackOff

x73

2026-03-03T08:44:40Z

Back-off restarting failed container heap-oom-prom in pod heap-oom-prom-67d785cb84-d7vj1_chaos-test(18f6bb00-524a-4f2d-88b8-0aa6d47782a9)

NORMAL Scheduled

x1

2026-03-03T08:48:52Z

Successfully assigned chaos-test/heap-oom-prom-67d785cb84-f8zfd to ip-10-0-31-80.us-east-2.compute.internal

NORMAL AddedInterface

x1

2026-03-03T08:48:53Z

Add eth0 [10.131.0.26/23] from ovn-kubernetes

NORMAL Pulling

x1

2026-03-03T08:48:53Z

Pulling image "quay.io/causa-ai-hub/quarkus-heap-oom:heap-oom-prom"

NORMAL Pulled

x1

2026-03-03T08:48:55Z

Successfully pulled image "quay.io/causa-ai-hub/quarkus-heap-oom:heap-oom-prom" in 2.231s (2.231s including waiting)

Logs

274 raw / 266 dedup

6 Errors 2 Warnings 253 Info

Raw Logs (274 lines)

Deduplicated Logs (266 lines)

Picked up JAVA_TOOL_OPTIONS: -XX:+UseContainerSupport -XX:MaxRAMPercentage=80 -XX:+CrashOnOutOfMemoryError

2026-03-03 09:46:59,746 WARN [io.micrometer.core.instrument.MeterRegistry] (main) This Gauge has been already registered (MeterId{name='netty.eventexecutor.workers', tags=[]}), the registration will be ignored. Note that subsequent logs will be logged at debug level.

2026-03-03 09:47:00,036 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-2) Scrape started. targets=0

2026-03-03 09:47:00,036 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-2) Scrape completed. processed=0 targets

2026-03-03 09:47:00,538 INFO [io.quarkus] (main) heap-oom-prom 1.0-SNAPSHOT on JVM (powered by Quarkus 3.32.1) started in 3.083s. Listening on: http://0.0.0.0:8080

2026-03-03 09:47:00,538 INFO [io.quarkus] (main) Profile prod activated.

2026-03-03 09:47:00,538 INFO [io.quarkus] (main) Installed features: [cdi, micrometer, resteasy-jackson, scheduler, smallrye-context-propagation, vertx]

2026-03-03 09:47:00,642 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-3) Writer started. targets=50

2026-03-03 09:47:00,868 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-3) Writer completed. wrote=50 entries

2026-03-03 09:47:00,868 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-3) Heap used=9 MB

2026-03-03 09:47:05,001 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-2) Writer started. targets=250

2026-03-03 09:47:05,001 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-1) Scrape started. targets=255

2026-03-03 09:47:05,003 INFO [ai.causa.scheduler.ScrapeScheduler] (executor-thread-1) Scrape completed. processed=255 targets

Model Information

ANOMALY DETECTION MODEL llama2:7b-chat-q8_0	RCA ANALYSIS MODEL llama2:7b-chat-q8_0	VALIDATION MODEL llama2:7b-chat-q8_0
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SESSION ID:
92725b47-f8ae-4d4f-b79b-f4d660a415e9

